

ToddySoft Connect ADS Driver - Ignition 8.3

Ignition 8.3 module for connecting to Beckhoff TwinCAT PLCs via the ADS/AMS protocol, with support for plain TCP and encrypted "Secure ADS" (TLS) connections.

The device configuration uses a custom configuration form that adapts to the selected **Transport**: only the settings relevant to the chosen transport are shown, so you are never presented with fields that do not apply.

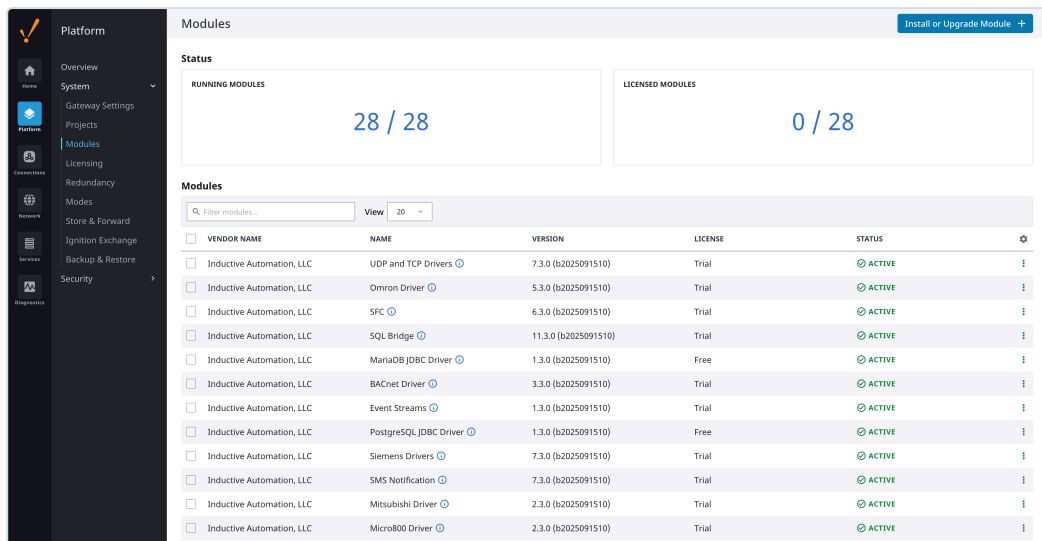
Installation

This module is for Ignition 8.3 only. It is built against the Ignition 8.3 SDK and will not run on any other major version. For Ignition 8.1, install the separate `...-8.1.modl` build instead — the two are not interchangeable, and each gateway needs the build that matches it.

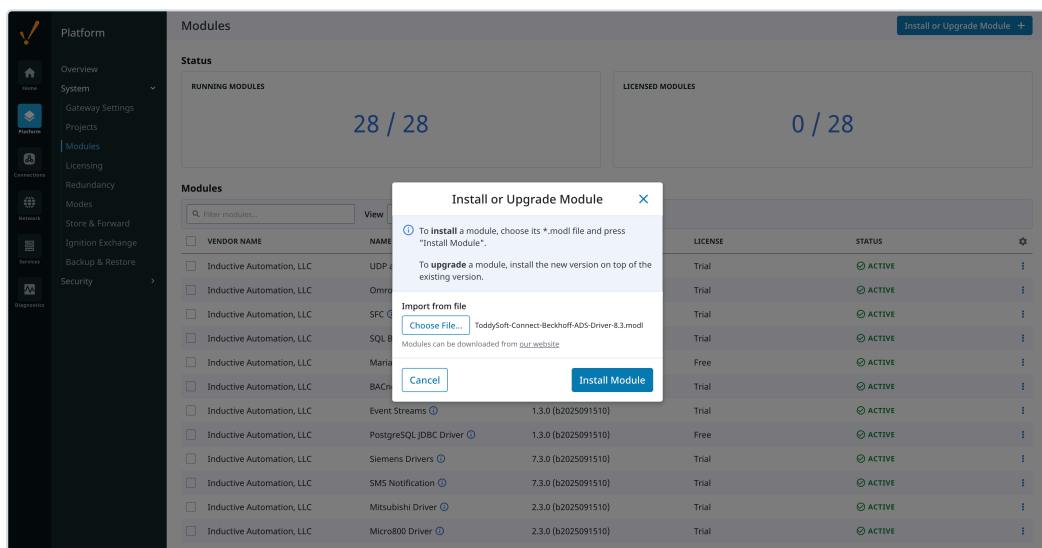
Ignition 8.0 and earlier are not supported and cannot be. The module is built on the device-driver API that Ignition introduced with 8.1, so on 8.0 the gateway fails to load it with `NoClassDefFoundError: ... opcu.server.api.AbstractDeviceModuleHook`.

Both builds carry the same name in the gateway's module list, so tell them apart by the file you installed or by the module description, which names the Ignition version.

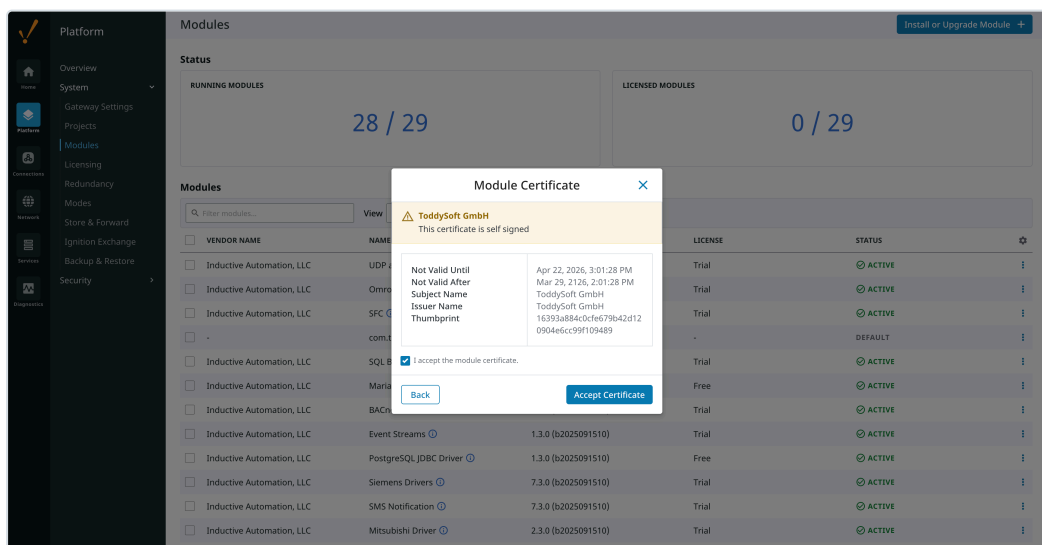
1. Open the Ignition Gateway web interface and navigate to **Config > System > Modules**.



2. Click **Install or Upgrade a Module...**



3. Select the **ToddySoft-Connect-Beckhoff-ADS-Driver-8.3.modl** file and upload it.



4. Accept the license / certificate prompt to continue the installation.

The screenshot shows the 'Modules' page in the Inductive Automation Platform. The left sidebar contains navigation links: Overview, System, Gateway Settings, Projects, Modules (selected), Licensing, Redundancy, Modes, Store & Forward, Ignition Exchange, Backup & Restore, and Security. The main content area is titled 'Modules' and features a 'Status' section with two boxes: 'RUNNING MODULES' showing '28 / 29' and 'LICENSED MODULES' showing '0 / 29'. Below this is a table of modules with columns: Vendor Name, Name, Version, License, and Status. The table lists 13 modules, all with a status of 'ACTIVE' except for 'ToddySoft Connect Beckhoff ADS D river' which is 'INACTIVE, PENDING RESTART'.

VENDOR NAME	NAME	VERSION	LICENSE	STATUS
Inductive Automation, LLC	UDP and TCP Drivers	7.3.0 (b2025091510)	Trial	ACTIVE
Inductive Automation, LLC	Omron Driver	5.3.0 (b2025091510)	Trial	ACTIVE
Inductive Automation, LLC	SFC	6.3.0 (b2025091510)	Trial	ACTIVE
ToddySoft GmbH	ToddySoft Connect Beckhoff ADS D river	1.0.0-SNAPSHOT (b0)	n/a	INACTIVE, PENDING RESTART
Inductive Automation, LLC	SQL Bridge	11.3.0 (b2025091510)	Trial	ACTIVE
Inductive Automation, LLC	MariaDB JDBC Driver	1.3.0 (b2025091510)	Free	ACTIVE
Inductive Automation, LLC	BAChet Driver	3.3.0 (b2025091510)	Trial	ACTIVE
Inductive Automation, LLC	Event Streams	1.3.0 (b2025091510)	Trial	ACTIVE
Inductive Automation, LLC	PostgreSQL JDBC Driver	1.3.0 (b2025091510)	Free	ACTIVE
Inductive Automation, LLC	Siemens Drivers	7.3.0 (b2025091510)	Trial	ACTIVE
Inductive Automation, LLC	SMS Notification	7.3.0 (b2025091510)	Trial	ACTIVE
Inductive Automation, LLC	Mitsubishi Driver	2.3.0 (b2025091510)	Trial	ACTIVE

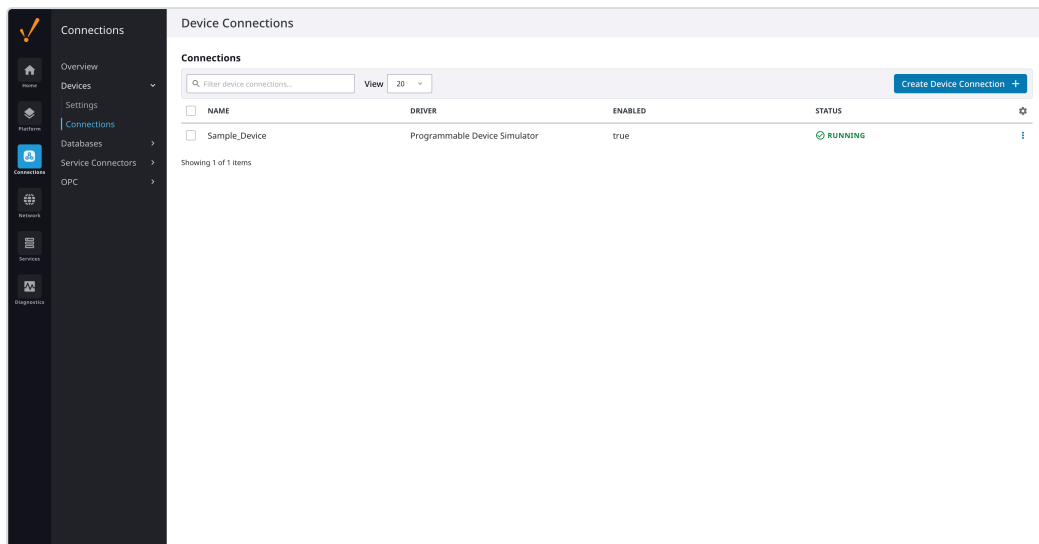
5. Wait for the Gateway to finish and verify the module appears as **Running** in the modules list.

The screenshot shows the 'Modules' page in the Inductive Automation Platform. The left sidebar is the same as in the previous screenshot. The main content area is titled 'Modules' and features a 'Status' section with two boxes: 'RUNNING MODULES' showing '29 / 29' and 'LICENSED MODULES' showing '0 / 29'. Below this is a table of modules with columns: Vendor Name, Name, Version, License, and Status. The table lists 13 modules, all with a status of 'ACTIVE'.

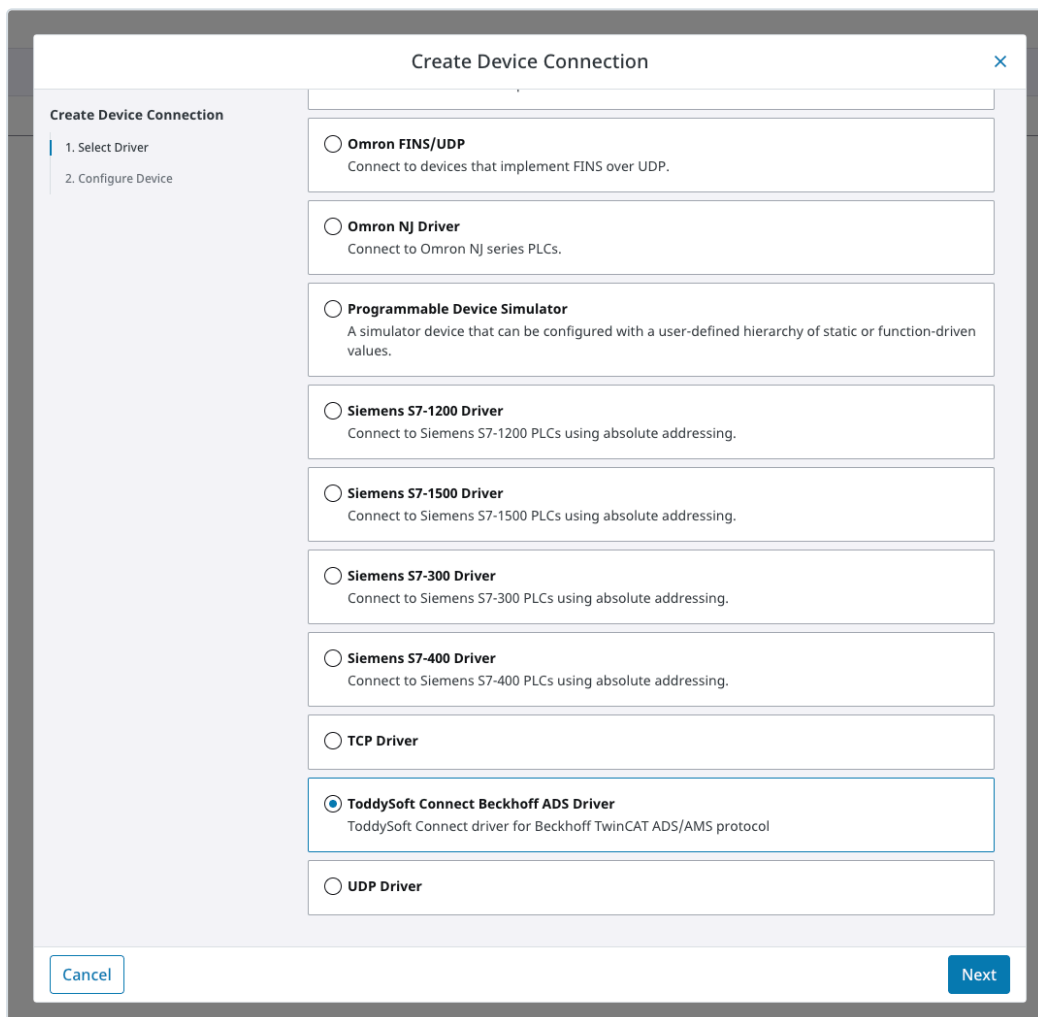
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Inductive Automation, LLC	BAChet Driver	3.3.0 (b2025091510)	Trial	ACTIVE
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Inductive Automation, LLC	SMS Notification	7.3.0 (b2025091510)	Trial	ACTIVE
Inductive Automation, LLC	Mitsubishi Driver	2.3.0 (b2025091510)	Trial	ACTIVE

Creating a Device Connection

1. Go to **Config > OPC UA > Device Connections**.



2. Click **Create new Device...** and select **ToddySoft Connect ADS Driver** as the device type.



3. Configure the connection settings (see the sections below) and click **Create Device Connection**.
4. The new device then appears in the Device Connections list.

NAME	DRIVER	ENABLED	STATUS
Sample_Device	Programmable Device Simulator	true	🟢 RUNNING
Simple-ADS	ToddySoft Connect Beckhoff ADS Driver	true	🟢 CONNECTED

Finding devices, and checking one before you save it

Above the configuration form sits a row of actions: **Discover Devices**, **Test Connection** and, once a check has found something to fix, the setup actions that repair it. They are at the top on purpose — the form is long, and none of this should require scrolling to the end of it.

Configure Device

GENERAL

*Required

Name*

A unique name for this device connection.

Description

An optional description for this device connection.

Enabled

☒ Set whether this device connection is enabled or disabled within the Gateway.

Discover Devices

Test Connection

Discover Devices

Sends an ADS broadcast on the gateway's own networks and lists the controllers that answer. Results are grouped by device, with one row per transport that device offers, so a controller reachable over plain AMS, Secure ADS and TLS-PSK appears once with three ways in.

Discover Devices

Devices that answered a broadcast on this gateway's networks.

Timeout (ms)

5000

Rescan

How long to wait for devices to answer. The gateway clamps this to its own bounds.

BTN-000ujn0t

192.168.24.50

tcp

AMS Net Id

192.168.24.50.1.1

tls

AMS Net Id

192.168.24.50.1.1

tls-psk

AMS Net Id

192.168.24.50.1.1

Windows-11-PLC

192.168.24.89

tcp

AMS Net Id

192.168.42.89.1.1

tls

AMS Net Id

192.168.42.89.1.1

tls-psk

AMS Net Id

192.168.42.89.1.1

CX_29A5B2

192.168.24.40

tcp

AMS Net Id

192.168.24.40.1.1

Windows-11-PLC

192.168.23.89

tcp

AMS Net Id

192.168.42.89.1.1

Cancel

Apply

Discovery does not work when Ignition runs inside Docker. The sweep is a UDP broadcast, and a container sits on the Docker-internal network rather than on the plant network the controllers are on, so the broadcast never reaches them and nothing answers. It is not a fault to diagnose — see [Docker related issues](#). Enter the device's host and AMS Net ID by hand; everything else on this page, **Test Connection** included, works normally.

Selecting a row fills in the form: host, port, target AMS Net ID and AMS port, and the transport itself. Three things it deliberately does not do:

- **It fills in only what the device actually reported.** A field the broadcast said nothing about is left alone rather than guessed at.
- **It never fills a password, key or keystore.** A broadcast tells you where a device is, never how to authenticate to it.
- **It does not fill in Source AMS Net ID or Source AMS Port.** Those describe *this gateway's* end of the connection, not the device, and the value a broadcast reports comes from whichever network interface happened to receive the reply — on a

gateway with several addresses, easily the wrong one. Left empty, the driver works them out from the connection it actually opens.

If applying a result would overwrite something you have already typed, it lists what would change and asks first. Running a sweep needs permission to create device connections; being logged in is not enough.

Test Connection

Asks the driver to diagnose the configuration exactly as entered — using the connection string the device itself would use, with secrets removed — and reports what it found rather than a bare success or failure. An unreachable device is reported as unknown rather than as broken, because those are different problems.

[Discover Devices](#) [Test Connection](#)

 **Setup required**
ads:tls://192.168.24.50:8016?target-ams-net-id=192.168.24.50.1.1&target-ams-port=851&tls.verify=false

Register a persistent AMS route on the target over Secure ADS, authenticating with a username and password inside the TLS session. — **SETUP_REQUIRED**
The AMS router answered on TCP and then refused the session. The most likely cause is that no AMS route is registered for this client — but a wrong target-ams-net-id, a wrong AMS port, an untrusted client certificate or rejected credentials all present identically. Observed: The target refused the Secure ADS session: UnknownCert. The connection targeted AMS port 851.

[Run](#)

Where the driver can repair what it found, the diagnosis carries the fix. The common case is a Secure ADS device whose AMS route has not been registered on the controller: the panel offers to register it, asking for the controller's administrator account. **Those credentials are used for that one call and are never stored** — not in the device configuration, not in the gateway's database.

Some fixes cannot be automated. A TLS-PSK route is one: TwinCAT will not accept it over the wire, and attempting it damages the route. In that case the panel prints the steps to do it by hand instead of offering a button that cannot work.

Registering an AMS route

A Beckhoff controller only answers a host it has an **AMS route** for. Until this release that route had to be created on the controller by hand — in TwinCAT XAE, or by editing `/etc/TwinCAT/3.1/Target/StaticRoutes.xml` — before an Ignition device could connect at all. The module can now do it for you.

When **Test Connection** finds that the route is missing, it offers **Register AMS route** on that finding. The dialog asks for:

Field	Notes
Route Name	The name the route is filed under on the controller. Pre-filled from this gateway's certificate, which is what the controller checks — usually leave it as it is.
Route Username	An administrator account on the controller , not an Ignition user. On a stock TwinCAT system this is Administrator .
Route Password	That account's password.

ads-route-tls



Register a persistent AMS route on the target over Secure ADS, authenticating with a username and password inside the TLS session.

Route Username

Account on the target permitted to add routes, typically 'Administrator'.

Route Password

Password for that account. Transmitted inside the TLS session.

Ignore Common Name

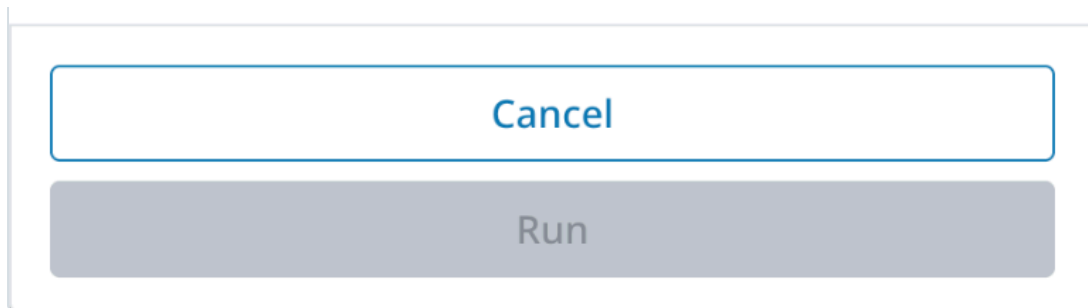
☐

Registers the route with the certificate common-name check waived, by setting the IGNORE_CN flag (bit 6) on the registration message — flags 0x00F0 instead of 0x00B0. The target records it as a property of the route: the entry in StaticRoutes.xml becomes `<Tls SelfSigned="true" IgnoreCn="true">`. Off by default, because it relaxes a check that exists to bind a route to one certificate identity; turn it on only when the announced machine name genuinely cannot be made to match the certificate's CN. Prefer setting the driver's machine-name parameter instead.

Route Name

Name the route is registered under on the target. Defaults to this machine's hostname, which is what an engineering tool would use.

Must match this gateway's certificate, whose common name is MacBook-Pro-von-Christofer-M2. The target checks the certificate, not the name you give the route.

A dialog box with a white background and a thin blue border. It contains two buttons stacked vertically. The top button is white with a blue border and the text 'Cancel' in blue. The bottom button is gray with a blue border and the text 'Run' in gray.

Those credentials are used for that single call and are **never stored** — not in the device configuration, not in the gateway's database, and they never appear in the connection string or the logs. If the route is later removed from the controller, you will be asked for them again.

Once the route is registered, run **Test Connection** again: the finding should clear and the device can be created.

Two limits worth knowing:

- **TLS-PSK routes cannot be registered this way.** TwinCAT will not accept a pre-shared key over the wire, and attempting it *empties the key on an existing route* — so the module refuses to try and shows the manual steps instead. Create a PSK route in TwinCAT.
- **A route is bound to a certificate.** If this gateway's certificate is replaced — including the automatic regeneration described under *Version history* — a route registered against the old one no longer matches and must be registered again.

A note on names, for Secure ADS with a self-signed certificate

Three names have to agree, and the certificate is the one that decides:

- the **common name** of this gateway's certificate,
- the **Machine Name** announced to the controller (blank means this gateway's host name, without its domain),
- the **name the route is filed under** on the controller.

If they disagree, the controller refuses the session with `CnMismatch` — after a perfectly good TLS handshake, on every connection. The module fills the route name in from the certificate for you when registering a route, and the **Machine Name** field is there for the case where the controller's route is already filed under something else.

Configuration

The configuration form is organized into the following sections, top to bottom:

General, Transport, Transport Settings, ADS / AMS Settings, Advanced Transport Settings and **Audit Log**.

General

Assigns a name (and optional description) to the connection and controls whether it is enabled.

GENERAL

*Required

Name *

Simple-ADS

The name of this device connection.

Description

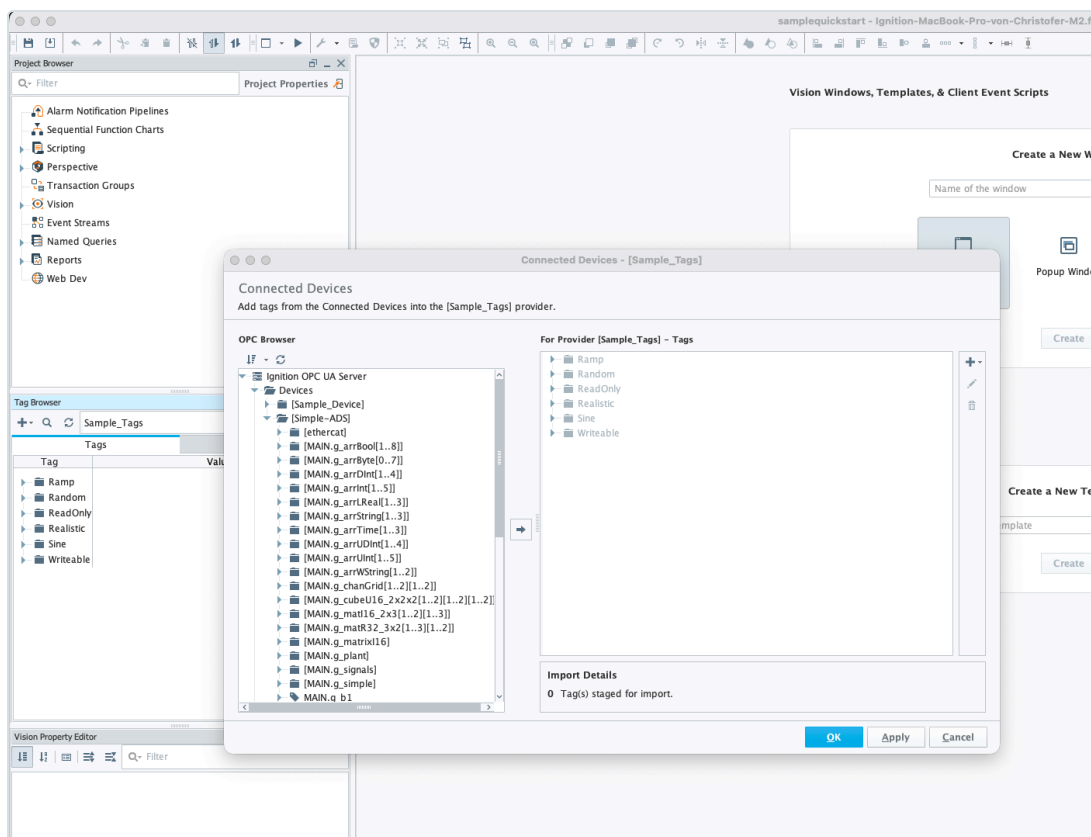
Device Connection using unencrypted ADS/AMS

Enabled

☒ Set whether this device connection is enabled or disabled within the Gateway.

Setting	Description	Default
Name	Unique name for this device connection	
Description	Optional description for this connection	
Enabled	Whether the connection is active	true

This is the name the connection is then referenced under in other parts of Ignition:



If **Enabled** is not checked, no connection is established until it is enabled.

Transport

Selects how the gateway connects to the ADS device. Choosing one of the TLS options reveals only the settings relevant to it in the **Transport Settings** section below.

TRANSPORT
*Required

Transport

TCP (plain, unsecured AMS)
▼

How the gateway connects to the ADS device. Choosing a TLS option reveals only the settings relevant to it.

Transport	Description
TCP (plain, unsecured AMS)	Plain, unencrypted ADS over TCP (default). No certificates or keys needed.
TLS-PSK	Encrypted ("Secure ADS") using a pre-shared key + identity. No certificates.
TLS-SSC (Self-Signed Certificate)	Encrypted using an automatically generated self-signed certificate.
TLS-SCA (Shared Certificate Authority)	Encrypted using certificates signed by a shared Certificate Authority (CA).

When any TLS transport is selected, the connection port automatically switches from 48898 to 8016 , unless a non-default port has been set explicitly.

The ADS route must already exist on the PLC. Earlier versions registered a route as part of connecting, which is why this page used to ask for a **Route Name**, **Route Username** and **Route Password**. Those three settings are gone. Registering a route is a one-time administrative act that needs the controller's administrator credentials, and requiring them here forced those credentials into the configuration that every reading device carries for the life of the connection.

Create the route on the controller before adding the device — in TwinCAT XAE, or by editing `/etc/TwinCAT/3.1/Target/StaticRoutes.xml` directly — and the device then connects with no administrator credentials at all. A device pointed at a controller that has no matching route will fail to connect.

Storing passwords and keys: The secret fields used by the TLS transports (`Keystore Password` and `Pre-Shared Key`) are handled through Ignition 8.3's secret management. For each you can choose **None**, store the value **Embedded** (encrypted within the device configuration) or **Referenced** from a configured secret provider. Values are never persisted in plain text.

Transport Settings

Always contains **Host** and **Port**; the remaining fields depend on the selected transport.

Setting	Description	Default
Host	Hostname or IP address of the TwinCAT device	localhost
Port	TCP port number (default 48898; TLS: 8016)	48898

TCP (plain, unsecured AMS)

Plain TCP shows only Host and Port — no security fields.

TRANSPORT SETTINGS

*Required

Host*

192.168.24.50

Hostname or IP address of the ADS device.

Port*

48898

ADS/AMS TCP port (default 48898; TLS transports use 8016).

TLS-PSK

Uses TLS-PSK, where both sides share a secret key and identity string. No certificates are needed — this is the simplest way to encrypt ADS communication.

TRANSPORT SETTINGS

*Required

Host*

192.168.24.50

Hostname or IP address of the ADS device.

Port*

8016

ADS/AMS TCP port (default 48898; TLS transports use 8016).

Identity*

MYPISKUSER

PSK identity name for TLS-PSK authentication.

Pre-Shared Key*

☐ None
 ☒ Embedded

Secret

.....

SHOW

☐ Referenced

Setting	Description	Default
Identity	PSK identity string (must match the PLC configuration)	(empty)
Pre-Shared Key	Hex-encoded shared secret key (secret field)	(None)

Configuring the PLC for PSK: the PLC must be configured with matching PSK settings. In the TwinCAT Secure ADS configuration (usually in `/etc/TwinCAT/3.1/Target/StaticRoutes.xml`):

- Set the **Identity** to match the value in Ignition (e.g., MYPISKUSER)
- Set the **Password/Key** that will be used to derive the pre-shared key

```

<Server>
  <Tls>
    <Psk>
      <Identity>MYPSKUSER</Identity>
      <Pwd>MySecret</Pwd>
    </Psk>
  </Tls>
</Server>
</RemoteConnections>
</TcConfig>

```

TwinCAT derives the PSK key by computing `SHA-256(Identity + Password)`. Note that the Identity must be **all uppercase**. To calculate the hex key needed for the Ignition configuration:

```
echo -n "<Identity><Password>" | shasum -a 256
```

For example, with Identity `MYPSKUSER` and Password `MySecret` :

```
echo -n "MYPSKUSERMySecret" | shasum -a 256
# Result:
a50d3154fea7bb54eec6710164e33ced872223f0a0ff05179e93e89e7f76c6f0
```

Use the resulting 64-character hex string as the **Pre-Shared Key** value in Ignition (stored Embedded or Referenced).

The corresponding route entry in `/etc/TwinCAT/3.1/Target/StaticRoutes.xml` looks like this:

```

<Route>
  <Name>MyDevMachine</Name>
  <Address>192.168.24.220</Address>
  <NetId>192.168.24.220.1.1</NetId>
  <Type>TCP_IP</Type>
  <Tls>
    <Psk>
      <Identity>MYPSKUSER</Identity>
      <Key>a50d3154fea7bb54eec6710164e33ced872223f0a0ff05179e93e89e7f76c6f0</Key>
    </Psk>
  </Tls>
</Route>

```

TLS-SSC (Self-Signed Certificate) and TLS-SCA (Shared Certificate Authority)

The two certificate-based transports share most of their fields. **TLS-SSC** uses an automatically generated self-signed certificate (the module manages the PKCS12 keystore for you; the ADS route itself must already exist on the controller). **TLS-SCA** uses certificates signed by a shared Certificate Authority that both Ignition and the PLC trust — the most secure option for production.

TRANSPORT SETTINGS

*Required

Host*

192.168.24.50

Hostname or IP address of the ADS device.

Port*

8016

ADS/AMS TCP port (default 48898; TLS transports use 8016).

Keystore Path*

/usr/local/bin/ignition/data/toddysoft/ads-client.p12

Path to the PKCS12 keystore file. Defaults to the module's auto-generated keystore if left empty.

Keystore Password

None

Embedded

Secret

SHOW

.....

Referenced

Ignore Common Name

Skip common name (CN) validation on the server certificate.

Setting	Description	Default
Keystore Path	Path to the PKCS12 keystore file. For TLS-SSC, leave empty to use the module's auto-generated keystore; for TLS-SCA, point to the CA-signed certificate.	<i>(auto / required)</i>
Keystore Password	Password for the keystore (secret field)	<i>(None)</i>
Machine Name	Name announced to the controller in the Secure ADS session. Leave empty to announce this gateway's host name without its domain. With a self-signed certificate it must equal the certificate's common name.	<i>(host name)</i>
Trust Store Path	<i>TLS-SCA only.</i> Path to a JKS/PKCS12 truststore holding the CA chain that signed the PLC's certificate	<i>(JVM defaults)</i>
Trust Store Password	<i>TLS-SCA only.</i> Password for the truststore (secret field)	<i>(None)</i>
Trust Store Type	<i>TLS-SCA only.</i> PKCS12 or JKS	<i>(from extension)</i>

The three **Trust Store** fields appear only when **TLS-SCA** is selected, because only that transport verifies the certificate the PLC presents. Note the direction: the **Keystore** holds *this gateway's own* certificate, which it presents to the PLC, while the **Trust Store** holds the CA certificate used to check the *PLC's* certificate.

Selecting **TLS-SCA** therefore shows three more fields than the form above:

TRANSPORT SETTINGS

*Required

Host*

192.168.24.50

Hostname or IP address of the ADS device.

Port*

8016

ADS/AMS TCP port (default 48898; TLS transports use 8016).

Keystore Path*

/usr/local/bin/ignition/data/toddysoft/ads-client.p12

Path to the PKCS12 keystore file. Defaults to the module's auto-generated keystore if left empty.

Keystore Password

☐ None

☒ Embedded

☐ Referenced

Secret

SHOW

.....

Ignore Common Name

☐ Skip common name (CN) validation on the server certificate.

Trust Store Path

/usr/local/bin/ignition/data/toddysoft/ads-ca.p12

Path to a JKS/PKCS12 truststore containing the CA chain that signed the device's certificate. Leave empty to validate against the JVM's default trust anchors.

Trust Store Password

☐ None

☒ Embedded

☐ Referenced

Secret

SHOW

.....

Trust Store Type

PKCS12

PKCS12 or JKS. Leave empty to infer it from the file extension.

If your CA is a private one (that is, not among the public authorities the Java runtime already trusts), set **Trust Store Path** to a truststore containing it. Left empty, the driver validates against the Java runtime's default trust anchors, which will reject a privately signed certificate.

When a keystore path is not specified (TLS-SSC), the module automatically generates a keystore in the Ignition module settings directory. The matching route in `/etc/TwinCAT/3.1/Target/StaticRoutes.xml` must already exist on the PLC.

ADS / AMS Settings

Configures the ADS-specific addressing. These settings apply regardless of the selected transport.

ADS / AMS SETTINGS

Target AMS Net ID

AMS Net ID of the target device, format x.x.x.x.x.x (6 octets).

Target AMS Port*
 
AMS port of the target (851 = TC3 runtime 1, 852/853 = runtime 2/3, 801 = TC2). Pick a common value or type a custom one.

Source AMS Net ID

AMS Net ID of this gateway, format x.x.x.x.x.x. Leave empty to derive automatically.

Source AMS Port

AMS port of this gateway (0 = auto-assign).

Request Timeout (ms)

ADS request timeout in milliseconds.

Browse Queries

Symbol browse query filter (e.g. MAIN.*).

Setting	Description	Default
Target AMS Net ID	AMS Net ID of the target device (format <code>x.x.x.x.x.x</code>). Defaults to <code>{target-ip}.1.1</code>	<i>(auto)</i>
Target AMS Port	AMS port of the target PLC runtime — pick a common TwinCAT port or type a custom value	<code>851</code>
Source AMS Net ID	AMS Net ID of this gateway (format <code>x.x.x.x.x.x</code>). Leave empty to derive automatically	<i>(auto)</i>
Source AMS Port	AMS port of this gateway (<code>0</code> = auto-assign)	<code>0</code>
Request Timeout	Default timeout for ADS requests in milliseconds	<code>4000</code>
Browse Queries	Symbol browse query filter. Separate multiple queries with <code>;</code> (e.g. <code>MAIN.*;GVL.*</code>)	<code>MAIN.*</code>

If not provided in the **Target AMS Net ID** field, the value is calculated by appending **.1.1** to the local IP address of the TCP connection. Note that there may be multiple paths to reach the PLC (if the device has multiple IP addresses). To control which local IP address is used, set the **Local Address** in the Advanced Transport Settings section.

The **Target AMS Port** field offers the common TwinCAT runtime ports as suggestions, each with a description, while still allowing any custom value:

Port	Description
851	TwinCAT 3 — PLC runtime 1 (default)
852 / 853	TwinCAT 3 — PLC runtime 2 / 3
801	TwinCAT 2 — PLC runtime 1
811 / 821 / 831	TwinCAT 2 — PLC runtime 2 / 3 / 4

Advanced Transport Settings

Lower-level TCP tuning options. These apply under whichever transport is active.

ADVANCED TRANSPORT SETTINGS

Connect Timeout (ms)

5000

Connection timeout in milliseconds (0 = default).

Read Timeout (ms)

0

Socket read timeout in milliseconds (0 = no timeout).

Write Timeout (ms)

0

Socket write timeout in milliseconds (0 = no timeout).

TCP No-Delay

☒ Enable TCP_NODELAY (disable Nagle's algorithm).

TCP Keep-Alive

☐ Enable TCP_KEEPALIVE.

Send Buffer Size

81920

Send buffer size in bytes.

Receive Buffer Size

81920

Receive buffer size in bytes.

Local Address

Select

Local address to bind to (blank = all interfaces).

Local Port

0

Local port to bind to (0 = ephemeral port).

Setting	Description	Default
Connect Timeout	Connection timeout in milliseconds (0 = default)	5000
Read Timeout	Socket read timeout in milliseconds (0 = no timeout)	0
Write Timeout	Socket write timeout in milliseconds (0 = no timeout)	0
TCP No-Delay	Enable TCP_NODELAY (disable Nagle's algorithm)	true
TCP Keep-Alive	Enable TCP_KEEPALIVE	false
Send Buffer Size	Send buffer size in bytes	81920

Setting	Description	Default
Receive Buffer Size	Receive buffer size in bytes	81920
Local Address	Local address to bind to, chosen from the gateway's network interfaces (blank = all interfaces)	(all)
Local Port	Local port to bind to (0 = ephemeral port)	0

The **Local Address** field is a dropdown populated with the gateway machine's local IPv4 addresses, which is especially useful when the host has multiple interfaces (see the Docker and VPN notes below).

Limits

Ceilings on what the controller can make the driver allocate. Each one bounds something the *other* side chooses - a declared table depth, a nesting depth, the size of a sum command - so a device that misbehaves cannot exhaust the gateway. The defaults are the driver's own; raise one only for a controller measured to need it. Leaving them alone produces exactly the connection the device made before these settings existed.

Setting	Description	Default
Max Browse Depth	How deep the browser dereferences a symbol into child nodes. A type declared in terms of another that refers back to it forms a cycle that parses fine and then walks forever.	20
Max Data-Type Table Depth	Nesting depth accepted when parsing the data-type table uploaded from the device. Bounds how deeply the table is <i>parsed</i> , where Max Browse Depth bounds how deeply it is <i>dereferenced</i> .	20
Max Sum Sub-Commands	Sub-commands packed into one Sum-Read or Sum-Write; a larger request is split across several. The default is Beckhoff's own recommendation - the PLC collects and copies every variable in the request before it can start its next cycle.	500
Max Sum Bytes	Bytes one Sum-Read or Sum-Write may move through the AMS router; a larger request is split. The default is the AMS router's own limit.	2097152

Audit Log

The **Audit Log** writes important diagnostic information to a dedicated log file. This file can grow quickly, so it is advised to only enable it when investigating issues.

AUDIT LOG

Audit Log File

Path to a file to which protocol audit events are written (leave empty to disable).

Setting	Description	Default
Audit Log File	Absolute path to the audit log file where events will be recorded. Leave empty to disable.	<i>(empty)</i>

NOTE: Please ensure the user the `Ignition` process is running under has write access to this path.

The Audit Log system automatically creates the file and rotates it (creating compressed files next to it) once a certain size is reached or on a time-based schedule.

Tag Addressing

Tags are addressed using symbolic names from the PLC program:

MAIN.bStart	# Boolean
MAIN.iCounter	# Integer
MAIN.fTemperature	# Float
GVL.sStatus	# String
MAIN.stMotor.fSpeed	# Struct member
MAIN.aValues[0]	# Array element
MAIN.stArray[1].nField	# Struct array element field

Configure **Browse Queries** (e.g., `MAIN.*`) to control which symbols appear when browsing tags in the Ignition Designer.

Tag Groups

By default, all tags are collected in the default tag group, which means all selected tags are fetched at the default interval of once a second.

If different intervals are desired, create additional tag groups, configure the desired intervals there, and assign tags to the corresponding tag groups.

Troubleshooting

Device shows "Faulted":

- Verify the PLC hostname/IP is reachable
- Check that AMS Net IDs match the TwinCAT configuration
- Ensure port 48898 (or 8016 for TLS) is not blocked by a firewall
- Review Gateway logs: **Status > Diagnostics > Logs**

Tags show bad quality:

- Verify the variable exists in the PLC program
- Check spelling and case sensitivity
- Ensure the PLC program is running

TLS connection fails:

- For TLS-PSK: verify the identity and key match on both sides
- For TLS-SSC: check that route credentials are correct
- For TLS-SCA: ensure both certificates are signed by the same CA

Docker related issues

When running `Ignition` inside `Docker`, the host receives an IP address on the Docker-internal network. When connecting to the PLC, the connection originates from one of the host's IP addresses. In this case the connection needs a **Source AMS Net ID** that refers to the host's IP address (and the **Local Address** in Advanced Transport Settings can be used to pin the originating interface).

Leave the **Source AMS Port** at `0`. The driver then adopts the number the local TCP port was assigned, which is different for every connection. A fixed value works as well, but only for a single device — see [Several devices on the same PLC](#).

Device discovery does not work from inside a container. *Discover Devices* broadcasts on the networks the gateway itself is attached to, which for a container are Docker's, not the plant's. No controller on the real network ever sees the request. Running the container on the host network (`--network host`, Linux only) puts the gateway on the plant network and discovery then works; a bridged container cannot be made to work by configuration. Type the host and AMS Net ID in by hand instead — that is the only part of the module a container costs you.

VPN related issues

When connecting to a remote PLC over a VPN, depending on the VPN type the host usually gets assigned a local VPN IP address. As with Docker, set a **Source AMS Net ID** that references this assigned IP address.

Leave the **Source AMS Port** at `0`. The driver then adopts the number the local TCP port was assigned, which is different for every connection. A fixed value works as well, but only for a single device — see [Several devices on the same PLC](#).

Several devices on the same PLC

The TwinCAT router routes by **Source AMS Net ID**. When two devices announce the same one to the same PLC, the router treats the second connection as a replacement for the first and closes it. The two devices then keep evicting each other, which shows up as connections that repeatedly drop and return to `Connection Lost` – see the `Gateway logs` for details.

Give every device that connects to the same PLC its own **Source AMS Net ID**, and add a matching route on the PLC for each of them. Leaving **Source AMS Net ID** empty does not avoid the clash: the driver then derives it from the gateway's IP address, which is the same for every device on that gateway.

The module detects the clash and reports it in the Gateway log when the device starts:

```
[Device B] Configuration conflict: source AMS Net ID
'192.168.1.100.1.1' towards 192.168.1.10
is also used by Device A. The PLC only accepts one connection per
identity and will keep closing
whichever connection it considers the older one, so these devices
will repeatedly disconnect each
other. Give each device its own identity.
```

Version history

Unreleased — 1.3.0

Built on ToddySoft Connect 1.3.0.

1.2.0 — 2026-09-11

Built on ToddySoft Connect 1.2.0.

- **Added: Discover Devices** — the device page can now broadcast on the gateway's networks and list the controllers that answer, grouped by device with a row per transport each one offers. Picking a result fills in host, port, AMS Net IDs and the transport; it fills only what the device actually reported, never a password, and asks first if that would overwrite something you had already typed. Requires permission to create device connections.
- **Added: Test Connection**, which asks the driver to diagnose the configuration as entered and says what is wrong rather than pass or fail. Where the driver can repair it, the diagnosis offers the fix: a Secure ADS device with no AMS route offers to register one, given the controller's administrator credentials — used for that one call and never stored. Where the fix cannot be automated, such as a TLS-PSK route, the steps to do it by hand are shown instead.

- **Added:** a **Machine Name** setting for the TLS transports. Secure ADS announces a machine name to the controller, and with a self-signed certificate it must equal the certificate's common name or the controller answers `CnMismatch`. Leave it empty to announce this gateway's host name without its domain.
- **Added:** the auto-generated keystore is now an explicit **Keystore** choice — *Auto-generated (module)* or *Custom path* — naming the file in use. A blank path always meant the module's own keystore; it simply looked unfilled. Nothing stored has changed.
- **Changed:** the **Ignore Common Name** setting has been removed. It never had any effect — the parameter behind it was read by nothing, because the ignore-CN flag exists only when *registering* a route, not when connecting. Nothing changes for a device that had it ticked. The same option now appears where it works, on the route registration offered by **Test Connection**, off by default.
- **Changed:** discovery no longer fills in **Source AMS Net ID** or **Source AMS Port**. Neither describes the device found — they describe this gateway's end, taken from whichever network interface happened to receive the broadcast reply, which on a gateway with several addresses is easily the wrong one and fails as a thirty-second timeout. Left empty, the driver derives both from the connection it actually opens.
- **Fixed:** Secure ADS with the auto-generated certificate is no longer refused by every TwinCAT controller. The certificate was issued to this gateway's fully qualified host name while Secure ADS announces the short one, so the controller compared the two and answered `CnMismatch` after a successful handshake. The certificate is now issued to the short name, and a gateway holding a stale one regenerates it at startup. **A route registered against the old certificate must be registered again.**
- **Fixed:** a TLS device using the auto-generated keystore could not be created at all. The form demanded a **Keystore Path**, but an empty path is exactly how the module's own keystore is chosen — so **Create Device Connection** stayed greyed out, with no message saying which field was at fault, while **Test Connection** on the same page reported success.
- **Changed:** a browsed tag now shows the name of the device it came from — `Wall-1000_MAIN.temp` rather than `MAIN.temp`, folders included. Ignition names a tag after the node it was dragged from, so without this two devices both offered `MAIN.temp` and dragging from each into one folder collided. Only the displayed name changes: the OPC item path behind every tag is untouched, so **existing tags keep working** and only newly created ones take the new name. An array element, which used to show as just `[0]`, now reads `Wall-1000_MAIN.g_arr[0]`.
- **Fixed:** a device whose PLC was switched off no longer reports **Connected**. The startup reachability check ran and its result was thrown away; with tags configured the first failed poll corrected the status within seconds, so this was only visible on a device with no tags, which polls nothing and sat claiming *Connected* indefinitely. The status is now honest from the start and the device reconnects on its own once

the PLC answers. A device with no tags is also checked every five minutes so an outage is noticed at all.

- **Fixed:** the individual nodes of a browsed array read the right element again. ToddySoft Connect 1.2.0 unified how arrays are written in an address, and an array's own bounds are now part of the address the driver reports; the module was still appending the element index to that, which asks for a second array dimension rather than for an element. An array declared with a lower bound other than zero — a TwinCAT `ARRAY [1..10]` — was also read one position out. Arrays with the usual zero lower bound were unaffected.
- **Fixed:** the **TLS-SCA (Shared Certificate Authority)** mode now presents this gateway's own certificate. It differs from TLS-SSC only in how the *PLC's* certificate is verified, so it was always meant to send the keystore — but only the self-signed mode ever did. A **Keystore Path** was collected, documented as required, and then never used, so a controller configured to demand a client certificate refused the connection. No configuration change is needed; a device already filled in starts working.
- **Fixed:** connecting over **TLS-PSK** no longer destroys the pre-shared key on the controller's route. Every Secure-ADS connect sent a route *registration* message carrying the PSK identity; TwinCAT took it as one, rewrote the route from a message whose format has no field for a key, and left the route's PSK empty — so an ordinary tag read wiped out the credential it had just authenticated with. The router keeps the route in memory, so nothing looked wrong until the controller was restarted, at which point the device could no longer connect and the route had to be entered again by hand. Fixed in ToddySoft Connect 1.2.0 and confirmed on hardware. If a route's PSK has already been emptied this way, re-enter it on the controller once; it will not be cleared again.
- **Changed (action required):** the **Route Name**, **Route Username** and **Route Password** settings have been removed. ToddySoft Connect 1.2.0 no longer registers an AMS route while connecting — that is now a separate, one-time administrative operation — so those three settings bound nothing and no route was ever registered from them. For a TLS/Secure-ADS device, create the route on the controller before it connects (TwinCAT XAE, or `/etc/TwinCAT/3.1/Target/StaticRoutes.xml`); a device pointed at a controller with no matching route will fail to connect. Since this release the module can register the route for you, from **Test Connection** on the device page — see *Registering an AMS route* above. Plain-TCP devices are unaffected.
- **Changed:** the keystore password and the pre-shared key are no longer written into the connection string. They are handed to the driver directly, so they no longer appear in the gateway log, in a support bundle, or in any string quoting the connection. Nothing changes on this page — the same fields, in the same places.
- **Added:** a **Limits** section with four ceilings on what the controller can make the driver allocate — browse depth, data-type table depth, and the two Sum-command sizes. The defaults are the driver's own, so an existing device connects exactly as before.

1.1.0

Built on ToddySoft Connect 1.1.0.

- **Fixed:** polling now recovers on its own after a long outage. A device whose PLC was switched off overnight, unreachable across a network outage, or idle while a licence was lapsed and later restored could keep failing every poll until the device was disabled and re-enabled by hand. The connection and tag polling are now rebuilt automatically once failures persist; existing OPC UA subscriptions are preserved.
- **Added:** three **Trust Store** settings, shown when the *TLS-SCA* transport is selected, so a PLC certificate signed by your own CA can be validated. Previously only certificate authorities already trusted by the Java runtime could be used. Leaving **Trust Store Path** empty keeps the previous behaviour.
- **Fixed:** ADS connection checks now use the configured AMS port, so a runtime on a non-default AMS port is no longer reported as unreachable while data still flows.

1.0.0

Initial release.